

15

2026

WEEK 51

DECEMBER

TUESDAY

ديسمبر

الثلاثاء

15

٦ كيهك ١٧٤٣

٦ رجب ١٤٤٨ هـ

STA TASK

$$\text{required time (Setup)} = T_{clk2} + T_{clk} - T_{setz} - T_{jitter}$$

$$= 9 + 25 - 4 - 2 = 28 \text{ ns}$$

Path 1

$$\text{Arrival time} = T_{clk1} + T_{q, max1} + T_{wire}$$

$$+ T_{comb2, max} = 6 + 6 + 2 + 13 = 27 \text{ ns}$$

Setup Slack = 1  $\rightarrow$  Met

Path 2

$$\text{Arrival time} = T_{clk1} + T_{q, max1} + T_{wire}$$

$$+ T_{comb1, max} = 6 + 6 + 5 + 12 = 29 \text{ ns}$$

evening

Setup Slack = -1  $\rightarrow$  Violate

notes



14

2026

WEEK 51

DECEMBER

MONDAY

ديسمبر

الاثنين

14

هـ كيهك ١٤٤٣ ق

رجب ١٤٤٨ هـ

Required Time (Hold) = 12 ns

Path 1

Arrival time =  $T_{clk1} + T_{qmin1} + T_{w}$   
 $+ T_{comb2,min} = 6 + 2 + 2 + 9 = 19 ns$

Path 2 Hold Slack = 1

Arrival time = 15 ns

Hold Slack = 3

evening

notes



13

2026

WEEK 51

DECEMBER

ديسمبر

SUNDAY

الأحد

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Methods to Solve Setup Violation.

① Cell upsizing      ③ reduce logic depth

② buffer inseration in CLK path

Methods to Solve hold Violation

① insert buffer in data path

② Cell downsizing

notes